

Metabolic Acidosis

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DEFINITION

Metabolic acidosis is defined as a low arterial blood pH in conjunction with a reduced serum bicarbonate concentration $[\text{HCO}_3^-]$. Respiratory compensation results in a decrease in arterial carbon dioxide tension (Paco_2). A low serum $[\text{HCO}_3^-]$ alone is not diagnostic of metabolic acidosis because it also results from the kidney compensation to chronic respiratory alkalosis. Measurement of the arterial pH differentiates between these two possibilities. Box 13.1 shows the expected compensatory responses for metabolic and respiratory acid-base disorders.¹

After the diagnosis of metabolic acidosis is confirmed, the first step in the examination of the patient is to calculate the serum anion gap. The *anion gap* equals the difference between the serum concentrations of the major cation sodium ($[\text{Na}^+]$) and the major measured anions chloride and bicarbonate ($[\text{Cl}^-]$ and $[\text{HCO}_3^-]$) and is given by the following formula:

$$\text{Anion gap} = [\text{Na}^+] - ([\text{Cl}^-] + [\text{HCO}_3^-])$$

In healthy individuals, the normal value of the anion gap is approximately 12 ± 2 mmol/L. Because many of the unmeasured anions consist of albumin, the normal anion gap is decreased by approximately 2.5 mmol/L for each 1 g/dL decrease in the serum albumin concentration below normal. The total number of cations must equal the total number of anions, so a decrease in the serum HCO_3^- concentration must be offset by an increase in the concentration of other anions. If the anion accompanying excess H^+ is Cl^- , the decrease in serum $[\text{HCO}_3^-]$ is matched by an equal increase in serum $[\text{Cl}^-]$. This acidosis is classified as a normal anion gap, a non-anion gap, or a hyperchloremic metabolic acidosis. In contrast, if excess H^+ is accompanied by an anion other than Cl^- , the decreased $[\text{HCO}_3^-]$ is balanced by an increase in the concentration of the unmeasured anion. $[\text{Cl}^-]$ remains the same. In this setting, the acidosis is said to be a high anion gap or anion gap metabolic acidosis.

The normal value for the anion gap has tended to fall over time because of changes in how serum Na^+ and Cl^- are measured.² Flame photometry for Na^+ measurement and a colorimetric assay for Cl^- have been replaced by the use of ion-selective electrodes, with which the serum Na^+ values have largely remained the same, whereas the serum Cl^- values have tended to be higher. As a result, the normal value for the anion gap has decreased to as low as 6 mmol/L in some reports. Recognizing this change, some laboratories have adjusted the calibration set point for Cl^- to return the normal value for the anion gap to the 12 ± 2 mmol/L range. The clinician needs to be aware that the average anion gap and range of normal values will vary across different facilities.

Fig. 13.1 provides a recommended approach to a patient with metabolic acidosis and lists the common causes of metabolic acidosis according to the anion gap.

NON-ANION GAP (NORMAL ANION GAP) METABOLIC ACIDOSIS

A non-anion gap metabolic acidosis can result from either kidney or extrarenal causes. Kidney causes of metabolic acidosis occur when renal bicarbonate generation, which results from net acid excretion, does not balance the loss of bicarbonate and other alkali buffers consumed in the buffering of normal endogenous acid production. This failure of net acid excretion is termed *renal tubular acidosis* (RTA). Extrarenal causes occur when exogenous acid loads, endogenous acid production, or endogenous bicarbonate losses are elevated and exceed kidney net acid excretion. The most common extrarenal cause of non-anion gap metabolic acidosis is chronic diarrhea.

Kidney and extrarenal causes of metabolic acidosis can be distinguished by measuring urinary ammonia excretion.³ The primary response of the kidney to metabolic acidosis is to increase urinary ammonia excretion, each millimole of urinary ammonia excreted resulting in the generation of 1 mmol of “new” bicarbonate. Thus, kidney causes of metabolic acidosis are characterized by low urinary ammonia excretion rates. In contrast, in extrarenal metabolic acidosis, urinary ammonia excretion is elevated. Because most laboratories do not measure urinary ammonia, one can indirectly assess ammonia excretion by measuring the *urinary anion gap* (UAG):

$$\text{UAG} = (\text{U}_{\text{Na}^+} + \text{U}_{\text{K}^+}) - \text{U}_{\text{Cl}^-}$$

The UAG is normally a positive value, ranging from +30 to +50 mmol/L. A negative value for the UAG suggests increased kidney excretion of an unmeasured cation (i.e., cation other than Na^+ or K^+). One such cation is NH_4^+ . With chronic metabolic acidosis because of extrarenal causes, urinary ammonia concentrations, in the form of NH_4Cl , can reach 200 to 300 mmol/L. As a result, the measured cation concentration will be less than the measured anion concentration, which includes the increased urinary Cl^- , and the UAG will be less than zero and frequently less than -20 mmol/L.

The UAG only indirectly reflects the urinary ammonia concentration and, if other unmeasured ions are excreted, can give misleading results. Examples include diabetic ketoacidosis, associated with substantial urinary excretion of sodium ketoacid salts, and toluene exposure (discussed later), associated with increased urinary excretion of sodium hippurate and sodium benzoate. In these settings, the UAG value may remain positive despite an appropriate increase in urinary ammonia excretion because of the increased urinary excretion of Na^+ acid-anion salts. A similar situation occurs when urinary NH_4^+ is excreted with an anion other than Cl^- , such as β -hydroxybutyrate, acetoacetate, bicarbonate, or hippurate. In these settings, and even when NH_4^+ is excreted with Cl^- , the urine osmolal gap (UOG) can be used as a surrogate for NH_4^+

BOX 13.1 Expected Compensatory Responses to Acid-Base Disorders

Acute Respiratory Acidosis

For every 10 mm Hg rise in P_{CO_2} , HCO_3^- increases by 1 mmol/L

Chronic Respiratory Acidosis

For every 10 mm Hg rise in P_{CO_2} , HCO_3^- increases by 3.5 mmol/L

Acute Respiratory Alkalosis

For every 10 mm Hg fall in P_{CO_2} , HCO_3^- decreases by 2 mmol/L

Chronic Respiratory Alkalosis

For every 10 mm Hg decrease in P_{CO_2} , HCO_3^- decreases by 5 mmol/L

Metabolic Acidosis

1.2 mm Hg decrease in P_{CO_2} for each 1 mmol/L fall in HCO_3^-

$$P_{CO_2} = HCO_3^- + 15$$

$$P_{CO_2} = \text{Last digits of pH}$$

Metabolic Alkalosis

P_{CO_2} increases by 0.7 for each mmol/L HCO_3^-

concentration. The UOG is the difference between the measured and the calculated urine osmolality (mOsmol/kg):

$$(2 \times [Na^+ + K^+]) + [Urea \text{ nitrogen (mg/dL)} / 2.8] + [Glucose \text{ (mg/dL)} / 18]$$

In SI units, the formula is:

$$2 \times [Na^+ + K^+] \text{ mmol/L} + Urea \text{ (mmol/L)} + Glucose \text{ (mmol/L)}$$

The normal value of the UOG is approximately 10 to 100 mOsmol/kg. NH_4^+ salts are generally the only other major urinary solute that contributes importantly to the urine osmolality, so values appreciably greater than 100 mOsmol/kg reflect increased excretion of NH_4^+ salts.

Urine pH, in contrast to the UAG or UOG, does not reliably differentiate acidosis of kidney origin from that of extrarenal origin. For example, an acid urine pH does not necessarily indicate an appropriate increase in net acid excretion. If renal ammonia metabolism is inhibited, as occurs with chronic hyperkalemia, there is decreased ammonia available in the distal nephron to serve as a buffer, and small amounts of distal H^+ secretion can lead to significant urine acidification. In this setting, the urine pH is acid, but net acid excretion is low because of the low ammonia excretion. Similarly, alkaline urine does not necessarily imply a renal acidification defect. In conditions in which ammonia metabolism is stimulated, distal H^+ secretion can be massive and yet the urine remains relatively alkaline because of the buffering effects of ammonia.

Metabolic Acidosis of Renal Origin

An overall approach to patient assessment for workup of metabolic acidosis of kidney origin is shown in Fig. 13.2.

Proximal Renal Tubular Acidosis (Type 2)

Normally, 80% to 90% of the filtered load of HCO_3^- is reabsorbed in the proximal tubule. In proximal (type 2) RTA, the proximal tubule has a decreased capacity to reabsorb filtered bicarbonate. When serum bicarbonate concentration is normal or nearly normal, the amount of bicarbonate filtered by the glomerulus exceeds proximal tubule bicarbonate reabsorptive capacity. When this happens, there is increased bicarbonate delivery to the loop of Henle and distal nephron that

Assessment of Low Serum HCO_3^- Concentration

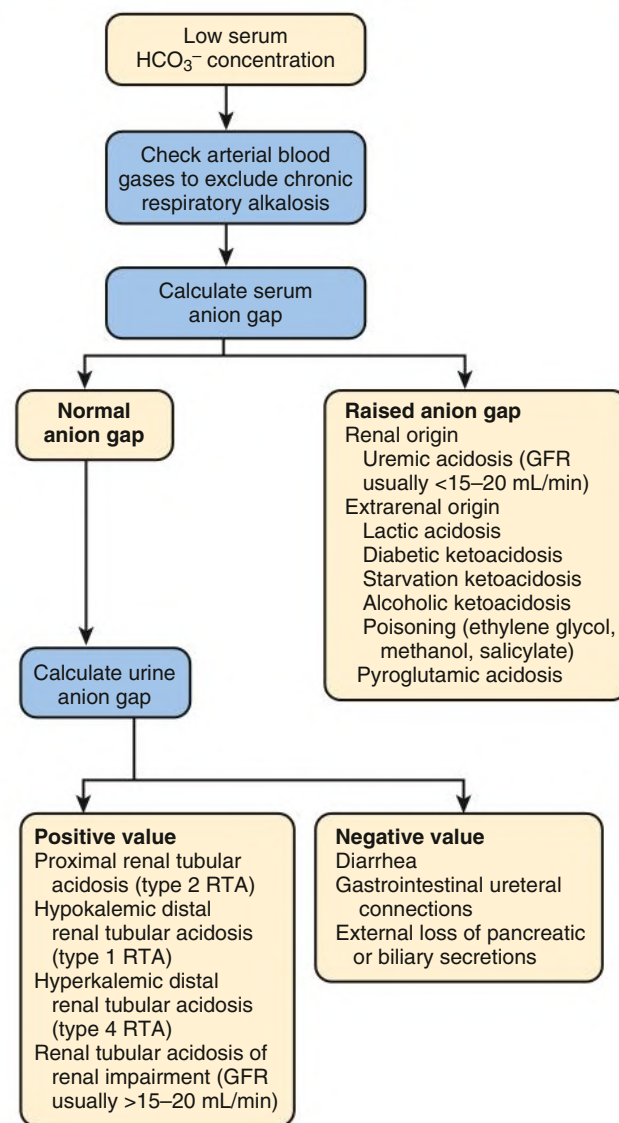


Fig. 13.1 Approach to the patient with low serum HCO_3^- concentration. *GFR*, Glomerular filtration rate; *RTA*, renal tubular acidosis.

exceeds their capacity to reabsorb bicarbonate. As a result, some filtered bicarbonate appears in the urine. The net effect is that serum $[HCO_3^-]$ decreases. Eventually, the filtered bicarbonate load decreases to the point at which the proximal tubule is able to reabsorb sufficient filtered bicarbonate that the bicarbonate load to the loop of Henle and the distal nephron is within their reabsorptive capacity. When this occurs, no further bicarbonate is lost in the urine, net acid excretion normalizes, and a new steady-state serum $[HCO_3^-]$ develops, although at a lower-than-normal level.

Hypokalemia is present in proximal RTA. Kidney $NaHCO_3$ losses lead to intravascular volume depletion, which in turn activates the renin-angiotensin-aldosterone system. Distal Na^+ delivery is increased as a result of the impaired proximal reabsorption of $NaHCO_3$. Because of the associated hyperaldosteronism and increased distal nephron Na^+ reabsorption, there is increased K^+ secretion. The net result is kidney potassium wasting and the development of hypokalemia. In the

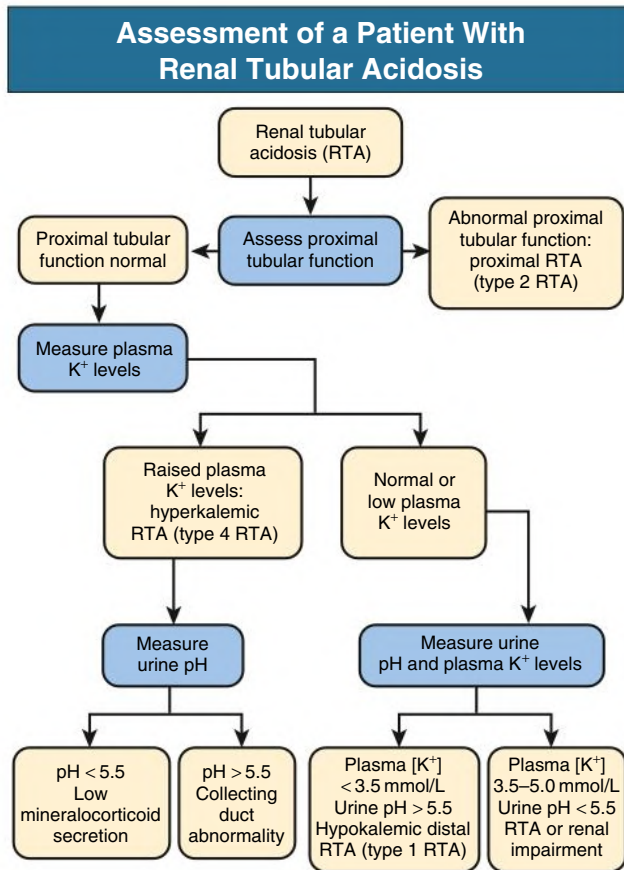


Fig. 13.2 Approach to the patient with renal tubular acidosis.

steady state, when virtually all the filtered HCO_3^- is reabsorbed in the proximal and distal nephron, kidney potassium wasting is less, and the degree of hypokalemia tends to be mild.

Proximal RTA may occur as an isolated defect in acidification but more commonly occurs in the setting of widespread proximal tubule dysfunction (Fanconi syndrome). In addition to decreased HCO_3^- reabsorption, patients with Fanconi syndrome have impaired reabsorption of glucose, phosphate, uric acid, amino acids, and low-molecular-weight proteins. Various inherited and acquired disorders have been associated with the development of Fanconi syndrome and proximal RTA (Box 13.2). The most common inherited cause in children is cystinosis (see Chapter 50). Most adults with Fanconi syndrome have an acquired condition that is related to an underlying dysproteinemic condition, such as multiple myeloma.

Skeletal abnormalities are common in these patients. Osteomalacia can develop from chronic hypophosphatemia caused by renal phosphate wasting if Fanconi syndrome is present. These patients also may have a deficiency in the active form of vitamin D because of an inability to convert 25-hydroxyvitamin D₃ to 1,25-dihydroxyvitamin D in the proximal tubule.

In contrast to distal RTA, proximal RTA is not associated with nephrolithiasis or nephrocalcinosis. One exception is the use of topiramate,^{4,5} an antiepileptic drug that is increasingly used to treat a variety of neurologic and metabolic disorders. The drug exerts an inhibitory effect on kidney carbonic anhydrase activity, resulting in a proximal acidification defect similar to that observed with acetazolamide. Topiramate also is associated with hypocitraturia, hypercalciuria, and elevated urine pH, leading to an increased risk for kidney stone disease.

Proximal RTA should be suspected in a patient with a normal anion gap acidosis and hypokalemia who has an intact ability to acidify the urine to below 5.5 while in a steady state.⁶ Proximal tubular dysfunction, such as euglycemic glycosuria, hypophosphatemia, hypouricemia, and mild proteinuria, helps support this diagnosis. The UAG is

BOX 13.2 Causes of Proximal (Type 2) Renal Tubular Acidosis

Not Associated With Fanconi Syndrome

- Sporadic
- Familial

Disorder of Carbonic Anhydrase

- Drugs: acetazolamide, sulfanilamide, topiramate
- Carbonic anhydrase II deficiency

Associated With Fanconi Syndrome

- Selective (no systemic disease present)
- Sporadic

Familial

- Autosomal recessive proximal RTA with ocular abnormalities: $\text{Na}^+\text{-HCO}_3^-$ cotransporter (NBCe1) defect
- Autosomal recessive proximal RTA with osteopetrosis and cerebral calcification: carbonic anhydrase II defect
 - Generalized (systemic disorder present)
- Genetic disorders
 - Cystinosis
 - Wilson disease
 - Hereditary fructose intolerance
 - Lowe syndrome
 - Metachromatic leukodystrophy

Dysproteinemic States

- Myeloma kidney
- Light chain deposition disease

Hyperparathyroidism

- Primary
- Secondary

Drugs and Toxins

- Tenofovir
- Outdated tetracycline
- Ifosfamide
- Gentamicin
- Streptozocin
- Lead
- Cadmium
- Mercury

Tubulointerstitial Disease

- Posttransplantation rejection
- Balkan nephropathy
- Medullary cystic disease

Others

- Bone fibroma
- Osteopetrosis
- Paroxysmal nocturnal hemoglobinuria

greater than zero, indicating the lack of increase in net acid excretion. The metabolic acidemia in the steady state is maintained by a low rate of NH_4^+ excretion. The lesion causing impaired reabsorption of HCO_3^- in the proximal tubule leads to a more alkaline intracellular pH exerting an inhibitory effect on glutamine metabolism and suppressing ammoniogenesis.

Treatment of proximal RTA is difficult. Administration of alkali increases serum $[\text{HCO}_3^-]$, which increases urinary bicarbonate losses and thereby minimizes subsequent increases in the serum $[\text{HCO}_3^-]$. Moreover, the increased distal sodium load, in combination with increased circulating plasma aldosterone, results in increased kidney potassium wasting and worsening hypokalemia. As a result, substantial amounts of alkali, often in the form of a potassium salt, such as potassium citrate, are required to prevent worsening hypokalemia. Children with proximal RTA should be aggressively treated to normalize their serum $[\text{HCO}_3^-]$ to minimize growth retardation. These children may require large amounts of alkali therapy, typically 5 to 15 mmol/kg/day.

Adults with proximal RTA are frequently not treated as aggressively as children are because of the lack of systemic metabolic abnormalities or bone disease. Many clinicians administer alkali therapy if serum $[\text{HCO}_3^-]$ is less than 18 mmol/L to prevent severe acidosis. Whether more aggressive therapy to normalize serum $[\text{HCO}_3^-]$ is beneficial remains unknown. However, the large amounts of alkali required, about 700 to 1000 mmol/day for a 70-kg individual, make this approach problematic.

Hypokalemic Distal Renal Tubular Acidosis (Type 1)

In contrast to proximal RTA, patients with distal RTA are unable to acidify their urine, either under basal conditions or in response to metabolic acidosis.^{7,8} Type 1 RTA results from a reduction in net H^+ secretion in the distal nephron and prevents urinary acidification, thereby minimizing titratable acid excretion and urinary ammonia excretion. As a result, these patients are unable to match net acid excretion to endogenous acid production, and acid accumulation ensues. The subsequent metabolic acidosis stimulates reabsorption of bone matrix to release the calcium alkali salts present in bone. During prolonged periods, this can result in progressive osteopenia in adults and in osteomalacia in children.

Distal RTA can be caused by either impaired H^+ secretion (secretory defect) or an abnormally permeable distal tubule, resulting in increased backleak of normally secreted H^+ (gradient defect); it may be genetic or acquired. Certain medications, especially amphotericin, result in increased backleak of protons across the apical plasma membrane, leading to a gradient defect form of distal RTA.

For patients with a secretory defect, the inability to acidify the urine below pH 5.5 results from abnormalities in any of the proteins involved in collecting duct H^+ secretion. Some patients may have an isolated defect in the $\text{H}^+-\text{K}^+-\text{ATPase}$ that impairs H^+ secretion and K^+ reabsorption.⁹ A defect confined to the vacuolar H^+-ATPase also results in kidney potassium wasting.¹⁰ The development of systemic acidosis tends to diminish net proximal fluid reabsorption with an increase in distal delivery, resulting in volume contraction and activation of the renin-aldosterone system. Increased distal Na^+ delivery coupled to increased circulating levels of aldosterone then leads to increased kidney K^+ secretion.¹¹ Defects in the basolateral anion exchanger (AE1) also can cause distal RTA. In this case the lack of basolateral HCO_3^- exit leads to intracellular alkalinization, which inhibits apical proton secretion.

Patients with distal RTA have low ammonia secretion rates. The decreased secretion is caused by the failure to trap ammonia in the tubular lumen of the collecting duct as a result of the inability to lower luminal fluid pH. In addition, there is often impaired medullary

transfer of ammonia because of interstitial disease. Interstitial disease is frequently present in such patients through an associated underlying disease or as a result of nephrocalcinosis or hypokalemia-induced interstitial fibrosis.

In contrast to proximal RTA, nephrolithiasis and nephrocalcinosis are common.¹² Urinary Ca^{2+} excretion is high secondary to acidosis-induced bone mineral dissolution. Luminal alkalinization also inhibits calcium reabsorption, resulting in further increases in urinary calcium excretion.¹³ Calcium phosphate solubility is also greatly lowered at alkaline pH, and calcium phosphate stone formation is accelerated. Stone formation is further enhanced as a result of low urinary citrate excretion. Citrate is metabolized to HCO_3^- , and its kidney reabsorption is stimulated by metabolic acidosis, thereby minimizing the severity of metabolic acidosis. Urinary citrate also chelates urinary calcium, decreasing ionized calcium concentrations. Accordingly, the decreased citrate excretion that occurs in chronic metabolic acidosis as a result to distal RTA further contributes to both nephrolithiasis and nephrocalcinosis.

Distal RTA may be a primary disorder, either idiopathic or inherited, but it most often occurs in association with a systemic disease, one of the most common of which is Sjögren syndrome (Box 13.3). Hypergammaglobulinemic states as well as drugs and toxins also may cause this disorder.

A common cause of acquired distal RTA is glue sniffing. Inhalation of toluene from the fumes of model glue, spray paint, and paint thinners can give rise to hypokalemic normal anion gap acidosis through multiple mechanisms. First, toluene inhibits collecting duct proton secretion. Second, metabolism of toluene produces the organic acids hippuric and benzoic acid. These are buffered by sodium bicarbonate, resulting in metabolic acidosis and the production of sodium hippurate and sodium benzoate. If plasma volume is normal, these salts are rapidly excreted in the urine, and a non-anion gap metabolic acidosis develops. If plasma volume is decreased, urinary excretion is limited, these salts accumulate, and an anion gap metabolic acidosis develops.

Distal RTA should be considered in all patients with a non-anion gap metabolic acidosis and hypokalemia who have an inability to lower the urine pH maximally. A urine pH above 5.5 in the patient with systemic acidosis suggests distal RTA, and a UAG value greater than zero or lack of an increase in the UOG is confirmatory. Depending on the duration of the distal RTA, the metabolic acidosis can be mild or very severe, with a serum $[\text{HCO}_3^-]$ as low as 10 mmol/L. Urinary potassium losses lead to the development of hypokalemia. Severe hypokalemia (<2.5 mmol/L) may result in musculoskeletal weakness and nephrogenic diabetes insipidus. The latter occurs because hypokalemia decreases aquaporin 2 (AQP2) expression in the collecting duct, thereby minimizing the ability to concentrate urine. An abdominal ultrasound scan or radiograph may reveal nephrocalcinosis.

In patients with minimal disturbances in blood pH and plasma $[\text{HCO}_3^-]$, a test of urinary acidification is required. Traditionally, such a test involved oral NH_4Cl administration to induce metabolic acidosis with assessment of the kidney response by serial measurement of urine pH. Many patients poorly tolerate NH_4Cl ingestion because of gastric irritation, nausea, and vomiting. An alternative way to test the capacity for distal acidification is to administer furosemide and the mineralocorticoid fludrocortisone simultaneously.¹⁴ The combination of both increased distal Na^+ delivery and mineralocorticoid effect will stimulate distal H^+ secretion by both an increase in the luminal electronegativity and a direct stimulatory effect on H^+ secretion. Unaffected individuals will have urine pH lowered below 5.5 with either maneuver.

Correction of the metabolic acidosis in distal RTA can be achieved by administration of alkali in an amount only slightly greater than daily acid production, usually 1 to 2 mmol/kg/day. In patients with

BOX 13.3 Causes of Hypokalemic Distal (Type 1) Renal Tubular Acidosis (RTA)

Primary

- Idiopathic
- Familial

Secondary

Autoimmune Disorders

- Hypergammaglobulinemia
- Sjögren syndrome
- Primary biliary cirrhosis
- Systemic lupus erythematosus

Genetic Diseases

- Autosomal dominant RTA: anion exchanger 1 defect
- Autosomal recessive RTA: H⁺-ATPase A4 subunit
- Autosomal recessive with progressive nerve deafness: H⁺-ATPase B1 subunit

Drugs and Toxins

- Amphotericin B
- Toluene

Disorders With Nephrocalcinosis

- Hyperparathyroidism
- Vitamin D intoxication
- Idiopathic hypercalciuria

Tubulointerstitial Disease

- Obstructive uropathy
- Renal transplantation

severe K⁺ deficits, correction of the acidosis with HCO₃⁻, particularly if it is done with sodium alkali salts such as NaHCO₃, can lower serum potassium concentration to dangerous levels. In this setting, potassium replacement should begin before the acidosis is corrected. In general, a combination of sodium alkali and potassium alkali is required for long-term treatment of distal RTA. For patients with recurrent kidney stone disease caused by distal RTA, treatment of the acidosis increases urinary citrate excretion, which slows the rate of further stone formation and may even lead to stone dissolution.

Hyperkalemic Distal Renal Tubular Acidosis (Type 4)

Type 4 RTA is characterized by distal nephron dysfunction, resulting in impaired kidney excretion of both H⁺ and K⁺ and causing hyperchloremic normal gap acidosis and hyperkalemia.¹⁵ (Note that the term *type 3 RTA* is most often applied to a rare autosomal recessive syndrome resulting from carbonic anhydrase II deficiency with features of both proximal and distal RTA.)

Type 4 RTA occurs most frequently with mild to moderate impairment in kidney function; however, the magnitude of hyperkalemia and acidosis is disproportionately severe for the observed glomerular filtration rate (GFR). Whereas hypokalemic distal (type 1) RTA is also a disorder of distal nephron acidification, type 4 is distinguished from type 1 RTA on the basis of several important characteristics (Table 13.1). Type 4 RTA is also a much more common form of RTA, particularly in adults.

Hyperkalemic distal RTA results from deficient circulating aldosterone or abnormal cortical collecting duct (CCD) function, or is related to hyperkalemia. In either case, a defect in distal H⁺ secretion develops. Impaired Na⁺ reabsorption by the principal cell leads to a decrease in

TABLE 13.1 Differentiation of Renal Tubular Acidosis Types

Factor	Type 1	Type 2	Type 4
Serum K ⁺	Low	Low	High
Kidney function	Normal or near normal	Normal or near normal	Stage 3, 4, or 5 chronic kidney disease
Urine pH during acidosis	High	Low	Low or high
Serum HCO ₃ ⁻ (mmol/L)	10–20	16–18	16–22
Urine citrate	Low	High	Low
Fanconi syndrome	No	May be present	No

the luminal electronegativity of the CCD, which impairs distal acidification as a result of the decrease in driving force for H⁺ secretion into the tubular lumen. The H⁺ secretion is further impaired in this segment as well as in the medullary collecting duct as a result of either the loss of the direct stimulatory effect of aldosterone on H⁺ secretion or an abnormality in the H⁺-secreting cell.

A consequence of the decrease in luminal electronegativity in the CCD is impaired kidney K⁺ excretion. In addition, a primary abnormality in CCD transport can also impair K⁺ secretion. The development of hyperkalemia adds to the defect in distal acidification by decreasing the amount of ammonia available to act as a urinary buffer. Some studies suggest that hyperkalemia itself, through its effects on ammonia metabolism, is the primary mechanism by which metabolic acidosis develops in type 4 RTA.

The etiology of type 4 RTA includes disorders associated with decreased circulating levels of aldosterone and conditions associated with impaired CCD function. The most common disease associated with type 4 RTA in adults is diabetes mellitus. In these patients, primary NaCl retention leads to volume expansion and suppression and atrophy of the renin-secreting juxtaglomerular apparatus. Several common drugs, such as nonsteroidal antiinflammatory agents (NSAIDs), angiotensin-converting enzyme (ACE) inhibitors, and high doses of heparin, as used for systemic anticoagulation, can lead to decreased mineralocorticoid synthesis. Impaired function of the CCD can be a feature of structural damage to the kidney, as in interstitial renal diseases such as sickle cell nephropathy, urinary tract obstruction, and lupus. CCD function also may be impaired from use of drugs such as amiloride, triamterene, spironolactone, and finerenone.¹⁶

Type 4 RTA should be suspected in a patient with a normal gap metabolic acidosis associated with hyperkalemia. The typical patient is in the fifth to seventh decade of life with a long-standing history of diabetes mellitus with a moderate reduction in the GFR. Plasma [HCO₃⁻] is usually 18 to 22 mmol/L and serum [K⁺] between 5.5 and 6.5 mmol/L. Most patients are asymptomatic; however, the hyperkalemia may occasionally be severe enough to cause muscle weakness or cardiac arrhythmias. The UOG is not increased and the UAG value is slightly positive, indicating minimal ammonia excretion in the urine. When the disorder is caused by a defect in mineralocorticoid activity, patients typically have urine pH below 5.5, reflecting a more severe defect in ammonia availability than occurs with a defect in H⁺ secretion (Fig. 13.3). In patients with structural damage to the collecting duct, the urine pH may be alkaline, reflecting both impaired H⁺ secretion and decreased urinary ammonia excretion.

Urine pH in Type 4 Renal Tubular Acidosis

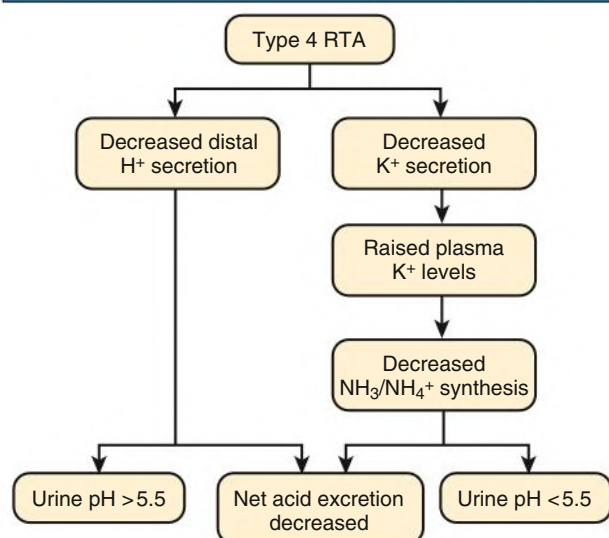


Fig. 13.3 Urine pH in Hyperkalemic Distal (Type 4) Renal Tubular Acidosis. Net acid excretion is always decreased; however, the urine pH can be variable. In structural disease of the kidney, the predominant defect is usually decreased distal H⁺ secretion and the urine pH is >5.5. In disorders associated with decreased mineralocorticoid activity, urine pH is usually <5.5.

Treatment of patients with type 4 RTA is directed at both the hyperkalemia and the metabolic acidosis. In many patients, lowering serum [K⁺] will simultaneously correct the acidosis.¹⁷ Correction of the hyperkalemia allows kidney ammonia production to increase, thereby increasing the buffer supply for distal acidification. The first consideration in treatment is to discontinue any nonessential medication that might interfere in either the synthesis or activity of aldosterone or the ability of the kidneys to excrete potassium (Box 13.4). ACE inhibitors and angiotensin receptor blockers usually should be continued because of the beneficial effects on cardiovascular disease and their renoprotective benefits in patients with chronic kidney disease (CKD). In patients with aldosterone deficiency who are neither hypertensive nor fluid overloaded, administration of a synthetic mineralocorticoid such as fludrocortisone 0.1 mg/day can be effective. In patients with hypertension or volume overload, particularly in association with CKD, administration of either a thiazide or a loop diuretic is frequently effective. Loop diuretics are required in patients with estimated GFR below 30 mL/min. Loop and thiazide diuretics increase distal Na⁺ delivery and thus stimulate K⁺ and H⁺ secretion in the collecting duct. Alkali therapy (e.g., NaHCO₃) also can be used to treat the acidosis and hyperkalemia, but the patient must be closely monitored to avoid volume overload and worsening hypertension. One can also consider chronic use of a gastrointestinal K⁺ binding drug such as patiomer or sodium zirconium cyclosilicate.¹⁸

Renal Tubular Acidosis in Chronic Kidney Disease

Metabolic acidosis in advanced CKD is caused by failure of the tubular acidification process to excrete the normal daily acid load. As functional kidney renal mass is reduced by disease, there is an adaptive increase in ammonia production and H⁺ secretion by the remaining nephrons. Despite increased production of ammonia from each remaining nephron, overall production may be decreased secondary to the decrease in total kidney mass. In addition, less ammonia is delivered to the medullary interstitium secondary to disrupted medullary anatomy.¹⁹ The ability to lower the urinary pH remains intact, reflecting the fact

BOX 13.4 Causes of Hyperkalemic Distal (Type 4) Renal Tubular Acidosis

Mineralocorticoid Deficiency Low Renin, Low Aldosterone

- Diabetes mellitus
- Drugs
 - NSAIDs
 - Cyclosporine, tacrolimus
 - β -Blockers

High Renin, Low Aldosterone

- Adrenal destruction
- Congenital enzyme defects
- Drugs
 - ACE inhibitors
 - ARBs
 - Heparin
 - Ketoconazole

Abnormal Cortical Collecting Duct

- Absent or defective mineralocorticoid receptor
- Drugs
 - Spironolactone, eplerenone
 - Triamterene
 - Amiloride
 - Trimethoprim
 - Pentamidine
- Chronic tubulointerstitial disease

ACE, Angiotensin-converting enzyme; ARB, angiotensin II receptor blocker; NSAIDs, nonsteroidal antiinflammatory drugs.

that the impairment in distal nephron H⁺ secretion is less than that in ammonia secretion. Quantitatively, the total amount of H⁺ secretion is small, and the acidic urine pH is the consequence of very little buffer in the urine. The lack of ammonia in the urine is reflected by a positive value for the UAG and lack of an increase in the UOG. Differentiation of RTA from type 4 RTA can be difficult because it is based on the clinician's determination of whether the severity of metabolic acidosis is out of proportion to the degree of kidney dysfunction.

Patients with CKD may develop a hyperchloremic normal gap metabolic acidosis associated with normokalemia or mild hyperkalemia as GFR decreases to less than 30 mL/min. With more advanced CKD (GFR <15 mL/min), the acidosis may change to an anion gap metabolic acidosis, reflecting a progressive inability to excrete phosphate, sulfate, and various organic acids. At this stage, the acidosis is commonly referred to as *uremic acidosis*.

Correction of the metabolic acidosis in patients with CKD is achieved by treatment with NaHCO₃, 0.5 to 1.5 mmol/kg/day, beginning when the HCO₃⁻ level is less than 22 mmol/L. In some patients, non-sodium citrate formulations can be used. Loop diuretics are often used in conjunction with alkali therapy to prevent volume overload. If the acidosis becomes refractory to medical therapy, dialysis needs to be initiated. Recent evidence suggests that metabolic acidosis in the patient with CKD should be aggressively treated because chronic acidosis is associated with metabolic bone disease, is associated with faster progression of kidney disease, and may lead to an accelerated catabolic state in patients with CKD.^{20,21}

Metabolic Acidosis of Extrarenal Origin

Diarrhea

Intestinal secretions from sites distal to the stomach are rich in HCO₃⁻. Accelerated loss of this HCO₃⁻ rich solution can result in metabolic

acidosis. The resultant volume loss signals the kidney to increase NaCl reabsorption; this combined with the intestinal NaHCO_3 losses generates a normal anion gap metabolic acidosis. The kidney response is to increase net acid excretion by increasing urinary excretion of ammonia.²² Hypokalemia, as a result of gastrointestinal losses, and the low serum pH both stimulate the synthesis of ammonia in the proximal tubule. The increase in availability of ammonia to act as a urinary buffer allows a maximal increase in H^+ secretion by the distal nephron.

The increase in urinary ammonia excretion associated with an extrarenal normal anion gap acidosis results in a negative UAG value and an increase in the UOG. Urine pH can be misleading and in chronic diarrhea may be above 6.0 because of substantial increases in renal ammonia metabolism that result in increased urine pH from the buffering ability of the ammonia. Although the clinical history should distinguish between these two possibilities, in a patient with surreptitious laxative abuse, this may not be helpful because diarrhea may not be reported. Colonoscopy may be required to demonstrate characteristic findings of laxative abuse (e.g., melanosis coli) if this diagnosis is being considered.

Treatment of diarrhea-associated metabolic acidosis is based on treatment of the underlying diarrhea. If this is not possible, alkali treatment, possibly including potassium alkali to treat hypokalemia and metabolic acidosis simultaneously, is indicated.

Ileal Conduits

Surgical diversion of the ureter into an ileal pouch is used in the treatment of the patient with neurogenic bladder or after cystectomy. The procedure may be associated rarely with development of a hyperchloremic normal anion gap metabolic acidosis. Acidosis in part is caused by reabsorption of urinary NH_4Cl by the intestine. The ammonia is transported through the portal circulation to the liver or is metabolized to urea to prevent hyperammonemic encephalopathy. This metabolic process consumes equimolar amounts of bicarbonate and therefore can result in the development of metabolic acidosis. Metabolic acidosis also may develop because urinary Cl^- can be exchanged for HCO_3^- through activation of a Cl^- - HCO_3^- exchanger on the intestinal lumen. In some patients, a kidney defect in acidification can develop and exacerbate the degree of acidosis. Such a defect may result from tubular damage caused by pyelonephritis or high colonic pressures, secondarily causing urinary obstruction.

The severity of acidosis relates to the length of time the urine is in contact with the bowel and the total surface area of bowel exposed to urine. In patients with a ureterosigmoid anastomosis, these factors are increased and the acidosis tends to be more common and more severe than in patients with an ileal conduit. The ileal conduit was designed to minimize the time and area of contact between urine and intestinal surface. Patients with surgical diversion of the ureter who develop metabolic acidosis should be evaluated for an ileal loop obstruction because this would lead to an increase in contact time between the urine and intestinal surface.

ANION GAP METABOLIC ACIDOSIS

Lactic Acidosis

Lactic acid is the end product in the anaerobic metabolism of glucose and is generated by the reversible reduction of pyruvic acid by lactic acid dehydrogenase and reduced nicotinamide adenine dinucleotide (NADH), as shown in the following formula:



Under normal conditions, the reaction is shifted toward the right and the normal lactate-to-pyruvate ratio is approximately 10:1. The

BOX 13.5 Causes of Lactic Acidosis

Type A (Tissue Underperfusion or Hypoxia)

Cardiogenic shock
Septic shock
Hemorrhagic shock
Acute hypoxia
Carbon monoxide poisoning
Anemia

Type B (Absence of Hypotension and Hypoxia)

Hereditary enzyme deficiency (glucose 6-phosphatase)

Drugs or toxins

- Phenformin, metformin
- Cyanide
- Salicylate, ethylene glycol, methanol
- Propylene glycol²⁶
- Linezolid²⁴
- Propofol²⁵
- Nucleoside reverse transcriptase inhibitors: stavudine, didanosine²⁴
- Clenbuterol²⁷
- Isoniazid

Thiamine deficiency

Systemic disease

- Liver failure
- Malignancy

reactants in this pathway are interrelated, as shown in the following equation:

$$\text{Lactate} = K[(\text{pyruvate})(\text{NADH})(\text{H}^+)]/\text{NAD}^+$$

where K is the equilibrium constant.

On the basis of this relationship, it is evident that lactate can increase for three reasons.²³ First, lactate can increase because of increased pyruvate production alone. In this situation, the normal 10:1 lactate-to-pyruvate ratio will be maintained. An isolated increase in pyruvate production can be seen in the setting of intravenous glucose infusions, intravenous administration of epinephrine, and respiratory alkalosis. Lactate levels in these conditions are minimally elevated, rarely exceeding 5 mmol/L. Second, lactate can increase as a result of an increased NADH:NAD⁺ ratio. Under these conditions, the lactate-to-pyruvate ratio can increase to very high values. Third, lactate can increase with a combination of increased pyruvate production and increased NADH:NAD⁺ ratio. This is common in severe lactic acidosis.

Lactic acidosis occurs when there is an imbalance between the production and the use of lactic acid. The net result is an accumulation of serum lactate and development of metabolic acidosis. The accumulation of the non-chloride anion lactate accounts for the increase in anion gap. Severe exercise and grand mal seizures are examples of lactic acidosis developing as a result of increased production. The short-lived nature of the acidosis in these conditions suggests that a concomitant defect in lactic acid use is present in most conditions of sustained and severe lactic acidosis.

Some of the disorders associated with the development of lactic acidosis are listed in Box 13.5. Type A lactic acidosis is characterized by underperfusion of tissue or acute hypoxia, such as hypotension, sepsis, acute tissue hypoperfusion, cardiopulmonary failure, severe anemia, hemorrhage, and carbon monoxide poisoning. Type B lactic acidosis occurs in the absence of overt hypoperfusion or hypoxia, such as with congenital defects in glucose or lactate metabolism, diabetes mellitus, liver disease, effects of drugs (especially overdose of isoniazid) and toxins, and neoplastic diseases.²⁴⁻²⁹

Thiamine deficiency is increasingly recognized as a cause of type B lactic acidosis in subjects with alcoholism, persistent vomiting, and severe malnutrition. In clinical practice, many patients will often exhibit features of type A and type B lactic acidosis simultaneously.

Therapy is aimed at correction of the underlying disorder. Restoration of tissue perfusion and oxygenation is attempted if these are compromised. The role of alkali in the treatment of patients with lactic acidosis is controversial; some experimental models and clinical observations suggest that administration of HCO_3^- may depress cardiac function and exacerbate the acidemia. In addition, such therapy may be complicated by volume overload, hypernatremia, and rebound alkalosis after the acidosis has resolved.³⁰ In general, HCO_3^- should be given when the systemic pH decreases to below 7.1 because hemodynamic instability becomes much more likely with severe acidemia. In such patients, alkali therapy should be directed at increasing the pH above 7.1; attempts to normalize the pH or $[\text{HCO}_3^-]$ should be avoided. Acute hemodialysis is rarely beneficial for lactic acidosis induced by tissue hypoperfusion. The hemodynamic instability that can occur with hemodialysis in these critically ill patients may worsen the underlying difficulty in tissue oxygenation.

Diabetic Ketoacidosis

Diabetic ketoacidosis results from the accumulation of acetoacetic acid and β -hydroxybutyric acid. The development of ketoacidosis is the result of insulin deficiency and a relative or absolute increase in glucagon.³¹ These hormonal changes lead to increased fatty acid mobilization from adipose tissue and alter the oxidative machinery of the liver such that delivered fatty acids are primarily metabolized into ketoacids. In addition, peripheral glucose use is impaired, and the gluconeogenic pathway in the liver is maximally stimulated. The resultant hyperglycemia causes an osmotic diuresis and volume depletion.

Ketoacidosis results when the rate of hepatic ketoacid generation exceeds renal excretion, causing increased blood ketoacid concentrations. The H^+ accumulation in the extracellular fluid (ECF) decreases HCO_3^- concentration, whereas the ketoacid anion concentration increases. An anion gap metabolic acidosis is the more common finding in the patient with diabetic ketoacidosis, but a normal gap metabolic acidosis also can be seen. In early stages of ketoacidosis, when the ECF volume is almost normal, ketoacid anions that are produced are rapidly excreted by the kidney as Na^+ and K^+ salts. Excretion of these salts is equivalent to the loss of potential HCO_3^- . This loss of potential HCO_3^- in the urine at the same time the kidney is retaining NaCl results in a normal gap metabolic acidosis. As volume depletion develops, kidney ketoacid excretion cannot match production rates and ketoacid anions are retained within the body, thus increasing the anion gap.³²

During treatment, the anion gap metabolic acidosis transforms once again into a normal gap acidosis. Treatment leads to a termination in ketoacid production. As the ECF volume is restored, there is increased kidney excretion of the Na^+ salts of the ketoacid anions. The loss of this potential HCO_3^- , combined with the retention of administered NaCl , accounts for the redevelopment of the hyperchloremic normal gap acidosis. In addition, K^+ and Na^+ administered in solutions containing NaCl and KCl enter cells in exchange for H^+ . The net effect is infusion of HCl into the ECF. The reversal of the hyperchloremic acidosis takes several days as the HCO_3^- deficit is corrected by the kidney.

Diabetic ketoacidosis can result in a severe metabolic acidosis with serum bicarbonate levels below 5 mmol/L. This diagnosis should be considered in patients with simultaneous metabolic acidosis and hyperglycemia. Diagnosis is confirmed by demonstration of retained ketoacids with nitroprusside tablets or reagent strips in the urine.

However, these tests detect only acetone and acetoacetate and not β -hydroxybutyrate. In the patient with lactic acidosis or alcoholic ketoacidosis, acetoacetate may be converted to β -hydroxybutyrate to an extent that depends on the NADH/NAD^+ ratio. With treatment of the diabetic ketoacidosis, acetoacetate is generated as this ratio falls, and the nitroprusside test result may suddenly become strongly positive.

The limitations of the nitroprusside test can be prevented by direct measurement of β -hydroxybutyrate. With uncontrolled diabetes, a serum β -hydroxybutyrate level above 3.0 mmol/L in adults and above 3.8 mmol/L in children confirms diabetic ketoacidosis.³³ Compared with urinary ketone measurements, capillary blood levels of β -hydroxybutyrate better correlate with both the degree of acidosis and the response to therapy.³⁴

Treatment consists of insulin and intravenous fluids to correct volume depletion. Deficiencies in K^+ , Mg^{2+} , and phosphate are common; therefore these electrolytes are typically added to intravenous solutions, especially after the initial fluid resuscitation. However, diabetic ketoacidosis typically manifests with hyperkalemia secondary to the insulin deficiency. Potassium should be administered only as hypokalemia develops, usually during insulin treatment of diabetic ketoacidosis. If there is significant hypokalemia at presentation, potassium supplementation may be needed before insulin administration to avoid life-threatening worsening of hypokalemia. Alkali therapy is generally not required because administration of insulin leads to the metabolic conversion of ketoacid anions to HCO_3^- and allows partial correction of the acidosis. However, HCO_3^- therapy may be indicated in patients who present with severe acidemia ($\text{pH} < 7.1$).³⁵

D-Lactic Acidosis

D-Lactic acidosis is a form of metabolic acidosis that can occur in the patient with small bowel resections or in patients with a jejunoileal bypass. Such short bowel syndromes create a situation in which carbohydrates that are normally extensively reabsorbed in the small intestine are delivered in large amounts to the colon. In the presence of colonic bacterial overgrowth, these substrates are metabolized into D-lactate and absorbed into the systemic circulation. Accumulation of D-lactate produces an anion gap metabolic acidosis in which the serum lactate concentration is normal because the standard test for lactate is specific for L-lactate. These patients typically present after ingestion of a large, high-carbohydrate meal, with neurologic abnormalities including confusion, slurred speech, and ataxia. Ingestion of low-carbohydrate meals and antimicrobial agents to decrease the degree of bacterial overgrowth are the principal treatments.

Starvation Ketosis

Abstinence from food can lead to a mild anion gap metabolic acidosis secondary to increased production of ketoacids. The pathogenesis of this disorder is similar to that of diabetic ketoacidosis in that starvation leads to relative insulin deficiency and glucagon excess. As a result, there is increased mobilization of fatty acids while the liver is set to oxidize fatty acids to ketoacids. With prolonged starvation, the blood ketoacid level can reach 5 to 6 mmol/L. The serum $[\text{HCO}_3^-]$ is rarely less than 18 mmol/L. More fulminant ketoacidosis is aborted because ketone bodies stimulate the pancreatic islets to release insulin and lipolysis is held in check. This break in the ketogenic process is notably absent in patients with insulin-dependent diabetes.³⁶ No specific therapy is indicated in starvation ketosis.

Alcoholic Ketoacidosis

Ketoacidosis develops in patients with a history of chronic ethanol abuse, decreased food intake, and often a history of nausea and vomiting. As with starvation ketosis, a decrease in the insulin-to-glucagon ratio leads

to accelerated fatty acid mobilization and alters the enzymatic machinery of the liver to favor ketoacid production. However, features unique to this disorder differentiate alcoholic ketoacidosis from simple starvation ketosis. First, the alcohol withdrawal combined with volume depletion and starvation greatly increases the levels of circulating catecholamines. As a result, the peripheral mobilization of fatty acids is much greater than typically found with starvation alone. This sometimes-massive mobilization of fatty acids can lead to marked ketoacid production and severe metabolic acidosis. Second, the metabolism of ethanol leads to accumulation of NADH. The increase in NADH/NAD⁺ is reflected by a higher ratio of β -hydroxybutyrate to acetoacetate. As mentioned previously, the nitroprusside reaction may be diminished by this redox shift despite the presence of severe ketoacidosis. Treatment of patients with alcoholic ketoacidosis focuses on glucose administration, which leads to the rapid resolution of the acidosis; stimulation of insulin release leads to diminished fatty acid mobilization from adipose tissue, as well as decreased hepatic output of ketoacids.³⁷

Ethylene Glycol and Methanol Intoxications

Ethylene glycol and methanol intoxications are characteristically associated with the development of a severe anion gap metabolic acidosis. Metabolism of ethylene glycol by alcohol dehydrogenase generates various acids, including glycolic, oxalic, and formic acids. Ethylene glycol is present in antifreeze and solvents and is ingested by accident or as a suicide attempt. The initial effects of intoxication are neurologic and begin with drunkenness but can quickly progress to seizures and coma. If left untreated, cardiopulmonary symptoms such as tachypnea, noncardiogenic pulmonary edema, and cardiovascular collapse may appear. From 24 to 48 hours after ingestion, patients may develop flank pain and acute kidney injury, often accompanied by abundant calcium oxalate crystals in the urine (Box 13.6). A fatal dose of ethylene glycol is approximately 100 mL.

Methanol is also metabolized by alcohol dehydrogenase and forms formaldehyde, which is then converted to formic acid. Methanol is found in a variety of commercial preparations, such as shellac, varnish, and deicing solutions, and is also known as wood alcohol. As with ethylene glycol, methanol can be ingested either by accident or as a suicide attempt. Clinically, methanol ingestion is associated with an acute inebriation followed by an asymptomatic period lasting 24 to 36 hours. Abdominal pain caused by pancreatitis, seizures, blindness, and coma may develop. The blindness is caused by direct toxicity of formic acid on the retina. Methanol intoxication is also associated with hemorrhage in the white matter and putamen, which can lead to the delayed onset of a parkinsonian syndrome (see Box 13.6). The lethal dose of methanol is 60 to 250 mL.

Lactic acidosis is also a feature of methanol and ethylene glycol poisoning and contributes to the elevated anion gap. Together with an elevated anion gap, an osmolar gap is an important clue to the diagnosis of ethylene glycol and methanol poisoning. The osmolar gap is the difference between the measured and calculated osmolality. The formula for the calculated osmolality is as follows:

$$2 \times \text{Na}^+ + \text{BUN}/2.8 + \text{Glucose}/18 + \text{EtOH}/4.6$$

where the blood urea nitrogen (BUN), glucose, and ethanol concentrations are in milligrams per deciliter. Inclusion of the ethanol concentration in this calculation is important because many patients ingest ethylene glycol or methanol while inebriated from ethanol ingestion. The normal value for the osmolar gap is less than 10 mOsm/kg. Each 100 mg/dL (161 mmol/L) of ethylene glycol will increase the osmolar gap by 16 mOsm/kg. Methanol contributes 32 mOsm/kg for each 100 mg/dL (312 mmol/L).

In addition to supportive measures, the patient with ethylene glycol and methanol poisoning is treated with fomepizole (4-methylpyrazole),

BOX 13.6 Ethylene Glycol and Methanol Poisoning

Time course of clinical symptoms and signs after ingestion

- Ethylene glycol
 - 0–12 hours: inebriation progressing to coma
 - 12–24 hours: tachypnea, noncardiogenic pulmonary edema
 - 24–36 hours: flank pain, renal failure, urinary calcium oxalate crystals
- Methanol
 - 0–12 hours: inebriation followed by asymptomatic period
 - 24–36 hours: pancreatitis, retinal edema progressing to blindness, seizures
 - >48 hours: putamen and white matter hemorrhage leading to Parkinson-like state

Increased anion gap metabolic acidosis

Increased osmolar gap

Treatment

- Supportive care
- Fomepizole (4-methylpyrazole) is agent of choice (competitor of alcohol dehydrogenase): 15 mg/kg IV loading dose, then 10 mg/kg every 12 hours for 48 hours. After 48 hours, increase dose to 15 mg/kg every 12 hours; increase frequency of dosing to every 4 hours during hemodialysis.
- Intravenous ethanol (5% or 10% solution) if fomepizole unavailable: Loading dose of 0.6 g/kg, followed by hourly maintenance dose of 66 mg/kg. Increase maintenance dose when the patient has history of chronic alcohol use and during hemodialysis.
- Hemodialysis to accelerate removal of parent compound and metabolites
- Bicarbonate therapy to treat acidosis

IV, Intravenous.

which inhibits alcohol dehydrogenase and prevents formation of toxic metabolites³⁸ (see Box 13.6). If fomepizole is unavailable, intravenous ethanol can be used to prevent the formation of toxic metabolites. Ethanol has more than a 10-fold greater affinity for alcohol dehydrogenase than that of other alcohols. Ethanol has its greatest efficacy when levels of 100 to 200 mg/dL are obtained. In addition to both fomepizole and ethanol therapy, the patient should receive hemodialysis to remove both the parent compound and its metabolites. Correction of the acidosis is accomplished with an HCO₃⁻-containing dialysate or by intravenous infusion of NaHCO₃.

Salicylate

Aspirin (acetylsalicylic acid) is associated with a large number of accidental or intentional poisonings. At toxic concentrations, salicylate uncouples oxidative phosphorylation and, as a result, leads to increased lactic acid production. In children, ketoacid production also may be increased. The accumulation of lactic, salicylic, keto, and other organic acids leads to the development of an anion gap metabolic acidosis. At the same time, salicylate has a direct stimulatory effect on the respiratory center. Increased ventilation lowers the carbon dioxide tension (Pco₂), contributing to the development of a respiratory alkalosis.³⁹ Children primarily manifest an anion gap metabolic acidosis with toxic salicylate levels; a respiratory alkalosis is most evident in adults.

In addition to conservative management, the initial goals of therapy for salicylate poisoning are to correct systemic acidemia and increase the urine pH. By increasing systemic pH, the ionized fraction of salicylic acid will increase, resulting in less accumulation of the drug in the central nervous system. Similarly, an alkaline urine pH favors increased urinary excretion because the ionized fraction of the drug is poorly reabsorbed by the tubule. At serum concentrations above 80

mg/dL or in the setting of severe clinical toxicity, hemodialysis can be used to accelerate drug elimination.

Pyroglutamic Acidosis

Pyroglutamic acid, also known as 5-oxoproline, is an intermediate in glutathione metabolism. An anion gap acidosis caused by pyroglutamic acid has been described rarely in critically ill patients receiving therapeutic doses of acetaminophen^{40,41} (Fig. 13.4). Patients present with severe anion gap metabolic acidosis accompanied by alterations in mental status ranging from confusion to coma. High concentrations of pyroglutamic acid are found in the blood and urine. Risk factors include malnutrition, infection, antibiotic use, pregnancy, and kidney failure. In these settings, glutathione levels are reduced because of the oxidative stress associated with critical illness. The reduction in glutathione secondarily leads to increased production of pyroglutamic acid. Even therapeutic doses of acetaminophen can precipitate this disorder

when risk factors are present. The diagnosis of pyroglutamic acidosis should be considered in patients with unexplained anion gap metabolic acidosis and recent acetaminophen ingestion.

ALKALI TREATMENT OF METABOLIC ACIDOSIS

Treatment of metabolic acidosis usually involves either sodium bicarbonate or citrate³⁵ (Table 13.2). NaHCO_3 can be taken orally as tablets or powder or given intravenously as a hypertonic bolus or an isotonic infusion, which can be created by adding 150 mmol NaHCO_3 to 1 L 5% dextrose in water (D5W). This solution is useful if treatment requires both volume expansion and alkali administration.

Citrate may be taken orally as a liquid, as sodium citrate, potassium citrate, or citric acid, or a combination. Many patients find citrate-containing solutions more palatable than oral NaHCO_3 as a source of oral alkali therapy. Oral citrate therapy should not be combined with medications that include aluminum. Citrate, which has a -3 charge under normal conditions, can complex with aluminum (Al^{3+}) in the intestinal tract, resulting in an uncharged moiety that is rapidly absorbed across the intestinal tract and then can dissociate to release free aluminum. This can increase the rate of aluminum absorption dramatically and in some patients, particularly those with severe CKD, has resulted in acute aluminum encephalopathy.

The dose of alkali therapy administered is based on both the total body bicarbonate deficit and the desired rapidity of treatment. Under normal circumstances, the volume of distribution (V_D) for bicarbonate is approximately 0.5 L/kg total body weight. Thus, the bicarbonate deficit, in millimoles, can be estimated from the following formula: $(0.5 \times \text{LBW}_{\text{kg}}) \times (24 - \text{HCO}_3^-)$, where LBW_{kg} is the lean body weight in kilograms and 24 is the desired resultant bicarbonate concentration.

Several caveats regarding this equation should be understood. First, edema fluid contributes to the volume of distribution of bicarbonate. Accordingly, an estimation of the amount of edema fluid should be included in this calculation. Second, the volume of distribution for bicarbonate increases as the severity of the metabolic acidosis worsens. When serum $[\text{HCO}_3^-]$ is 5 mmol/L or less, the volume of distribution may increase to 1 L/kg or more.

When acute treatment is desired, 50% of the bicarbonate deficit should be replaced during the first 24 hours. If hypertonic NaHCO_3 is administered, the increase in serum $[\text{HCO}_3^-]$ will be mirrored by an increase in serum $[\text{Na}^+]$. After the initial 24 hours of therapy, the response to therapy and the patient's current clinical condition are reevaluated before future therapy is decided. Acute hemodialysis solely for the treatment of metabolic acidosis, other than that associated with kidney failure, is rarely beneficial.

Mechanism of Pyroglutamic Acidosis

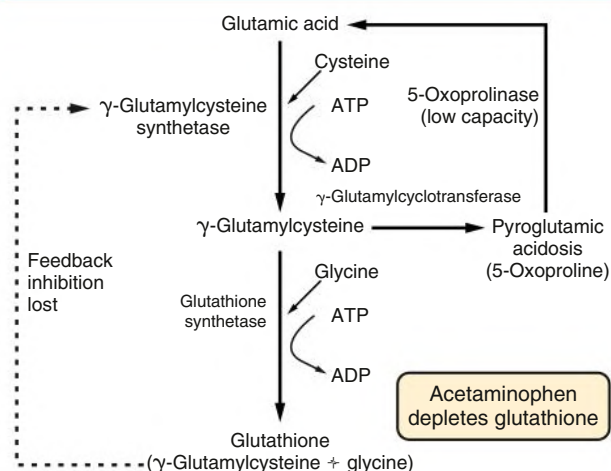


Fig. 13.4 Mechanism of Pyroglutamic Acidosis. Glutathione is formed from γ -glutamylcystine and glycine in the presence of glutathione synthetase. Glutathione normally regulates the activity of γ -glutamylcystine synthetase through feedback inhibition. Depletion of glutathione results in increased formation of γ -glutamylcystine, which in turn is metabolized to pyroglutamic acid (5-oxoproline) and cystine through γ -glutamylcyclotransferase. Pyroglutamic acid accumulates because the enzyme responsible for its metabolism (5-oxoprolinase) is low capacity. ADP, Adenosine diphosphate; ATP, adenosine triphosphate.

TABLE 13.2 Alkali Treatment Options

Therapy	Route	Usual Dose per Unit	Comments
Sodium bicarbonate tablet	PO	650 mg = 8 mmol	May cause gastric gas
Sodium bicarbonate	IV	50 mmol in 50 mL	Hypertonic, may cause hyponatremia
5% Dextrose with water with NaHCO_3	IV	150 mmol/L	Useful for simultaneous intravascular volume expansion and alkali administration
Potassium citrate (tablet)	PO	5 and 10 mmol per tablet	Useful for simultaneous K^+ and alkali therapy
Citric acid/potassium citrate/sodium citrate (liquid)	PO	1 mmol of Na^+ and K^+ and 2 mmol of citrate per milliliter	Avoid concomitant aluminum-containing medications such as antacids and sucralfate 1 mmol citrate equivalent to 3 mmol HCO_3^-
Potassium citrate (liquid)	PO	2 mmol of K^+ and 2 mmol of citrate per milliliter	Avoid concomitant aluminum-containing medications

IV, intravenous; PO, oral.

SELF-ASSESSMENT QUESTIONS

1. A 65-year-old man presents with the chief complaint of progressive weakness over the last several months. He is normotensive; physical examination is unremarkable with the following laboratory results (in mmol/L): Na^+ 135, Cl^- 105, K^+ 3.0, HCO_3^- 18, creatinine 1.8, blood urea nitrogen, glucose 110, Pco_2 28 mm Hg, pH 7.33, hematocrit 25%, white blood cells 5600/mm³, platelets 340,000/mm³, and urinalysis: trace protein, 1+ glucose, normal sediment, and 4.8 protein g/24 h. Which of the following is characteristic of the renal lesion present in this patient?
 - A. Nephrocalcinosis will be shown on a kidney, ureter, bladder radiograph of the abdomen.
 - B. Serum HCO_3^- concentration will increase after oral bicarbonate administration but then decrease to 18 mmol/L after therapy is discontinued.
 - C. Bicarbonate therapy will cause the serum K^+ to decline slightly because of a shift into cells.
 - D. Urine pH will be persistently alkaline.
 - E. Urine anion gap will be negative.
2. A 35-year-old woman presents having ingested a large quantity of aspirin after breaking up with her boyfriend. Her medical history is unremarkable, and she is taking no medications. Physical examination shows blood pressure 130/80 mm Hg lying and 110/62 mm Hg standing, pulse 102 beats/min, and respiratory rate 24 breaths/min; the remainder of the examination is normal. Laboratory tests on admission (in mmol/L) are Na^+ 138, K^+ 3.2, Cl^- 100, HCO_3^- 13, pH 7.48, and Pco_2 21 mm Hg; urinalysis is 1+ ketone, normal sediment; and urine chemistry (in mmol/L) is Na^+ 38, Cl^- <10, and K^+ 45. Which of the following statements is *true* regarding salicylate poisoning?
 - A. A 5-year-old child who has accidentally swallowed an unknown quantity of his mother's aspirin is likely to present with similar acid-base findings.
 - B. The cause of hypokalemia is primarily due to a shift of K^+ into cells because of respiratory alkalosis.
 - C. Serum uric acid levels are likely to be low in this patient.
 - D. The high urine Na^+ is caused by proximal tubular dysfunction as a result of salicylate nephrotoxicity.
 - E. Measurement of serum lactic acid in this patient is likely to be normal.
3. A 65-year-old White man with an extensive smoking history and known chronic obstructive pulmonary disease is admitted to the hospital with hematemesis. On the day of admission, endoscopy shows a nonbleeding duodenal ulcer. The procedure was complicated by the development of aspiration pneumonia and respiratory failure. A continuous infusion of intravenous lorazepam was required to control agitation and minimize peak inspiratory pressures. Admission laboratory results (in mmol/L) were Na^+ 142, K^+ 4.3, Cl^- 105, HCO_3^- 22, creatinine 1.4 mg/dL, blood urea nitrogen (BUN) 25 mg/dL, hematocrit 36, pH 7.35, Pco_2 45, and Po_2 75. On hospital day 4, gastrointestinal bleeding recurred, and the patient was taken to the operating room and underwent oversewing of a bleeding duodenal ulcer. The patient remained hemodynamically stable throughout the hospitalization but continued to require intravenous lorazepam for sedation. Laboratory results (in mmol/L) were Na^+ 138, K^+ 4.8, Cl^- 100, HCO_3^- 10, creatinine 1.8 mg/dL, BUN 28 mg/dL, glucose 120 mg/dL, and serum osmolality 330 mOsm/l (osmolar gap = 37). Which of the following is the cause of this patient's anion gap metabolic acidosis and increased osmolar gap?
 - A. Propylene glycol toxicity
 - B. Diabetic ketoacidosis
 - C. Lactic acidosis
 - D. Uremic acidosis
 - E. Isopropyl alcohol administration

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